Quizrr Chapter-Wise Test for JEE Advanced - 2025

By: C I P H Ξ R

Q.1

If the slope of Z (compressibility factor) v/s P curve is constant $\left(\text{ slope } = \frac{\pi}{492.6}\ \text{atm}^{-1}\right)$ at a particular temperature $(300\ \text{K})$ and very high pressure, then calculate diameter of the molecules. (Given : $\text{N}_\text{A} = 6.02 \times 10^{23}, \text{R} = 0.0821\ \text{atm. lit}\ \text{mol}^{-1}\ \text{K}^{-1}$)

(A) $7.5\mathring{A}$

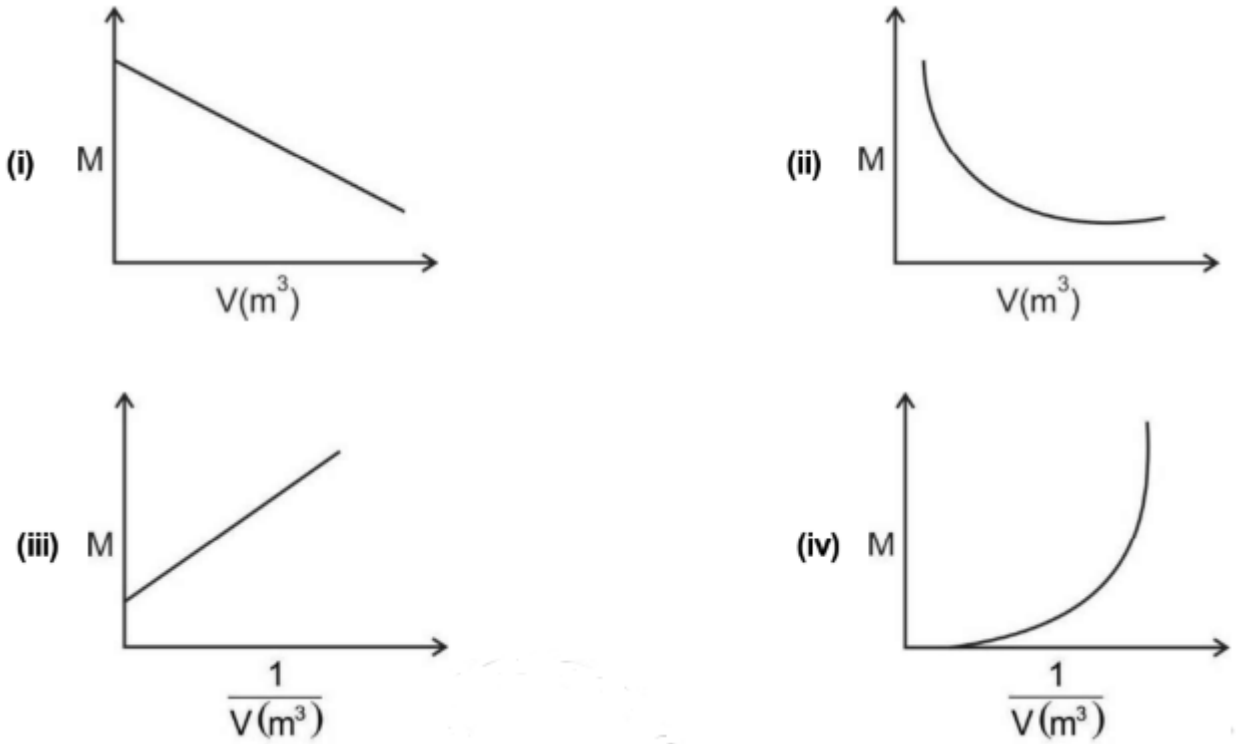
(B) $5\mathring{A}$

(C) $2.5\mathring{A}$

(D) $1.25\mathring{A}$

Q.2

Correct representation of graph between molarity and volume of solution in m^3 with a given amount of solute.



(A) (ii) and (iii)

(B) (i), (ii) and (iii)

(C) Only (ii)

(D) (i), (ii), (iii) and (iv)

Q.3

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Two identical bulbs containing ideal gases A and B are taken. Density of A is twice that of B. Mol wt. of A is half that of B. If the two gases are at the same temperature, the ratio of pressure of $A \& B$ is

(A) $1:2$

(B) $1:4$

(C) $4:1$

(D) $2:1$

Q.4

Pure O_2 diffuses through an aperture in 224 seconds, whereas mixture of O_2 and another gas containing $80\% O_2$ diffuses from the same in $234 \mathrm{sec}$. under similar condition of pressure and temperature. What is molecular wt. of gas?

(A) 23.2

(B) 46.6

(C) 18.4

(D) 32.2

Q.5

A car tyre has a volume of 10 litre when inflated. The tyre is inflated to a pressure of 3 atm at $17^{\circ} \mathrm{C}$ with air. Due to driving the temperature of the tyre increases to $47^{\circ} \mathrm{C}$. How many litres of air measured at $47^{\circ} \mathrm{C}$ and pressure of $1 \mathrm{~atm}$ should be let out to restore the tyre to 3 atm at $47^{\circ} \mathrm{C}$?

(A) 3.1 litre

(B) 6.2 litre

(C) 1.55 litre

(D) 9.3 litre

Q.6

Gaseous benzene reacts with hydrogen gas in presence of a platinum catalyst to form gaseous cyclohexane according to the reaction:

$$\mathrm{C_6H_6(\ g) + 3H_2(\ g) \rightarrow C_6H_{12}(\ g)}$$

A mixture of $\mathrm{C_6H_6}$ and excess $\mathrm{H_2}$ has a pressure of $80 \mathrm{~mm}$. of Hg. After the gas had been passed over a platinum catalyst and all the benzene converted to cyclohexane, the pressure of the gas was $40 \mathrm{~mm}$ of Hg under the same conditions. The volume fraction of $\mathrm{C_6H_6}$ present in the original mixture was:

(A) $1/3$

(B) $1/6$

(C) $1/5$

(D) $1/4$

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Q.7

A half litre container at $27^{\circ}\mathrm{C}$ is filled with $\mathrm{C_2H_4}$ at $1\ \mathrm{atm}$ and oxygen at $5\ \mathrm{atm}$ and the mixture is exploded. Calculate the final pressure if final temperature is $27^{\circ}\mathrm{C}$. The aqueous tension at $27^{\circ}\mathrm{C}$ is $38\ \mathrm{mm}$.

(A) $4.05\ \mathrm{atm}$.

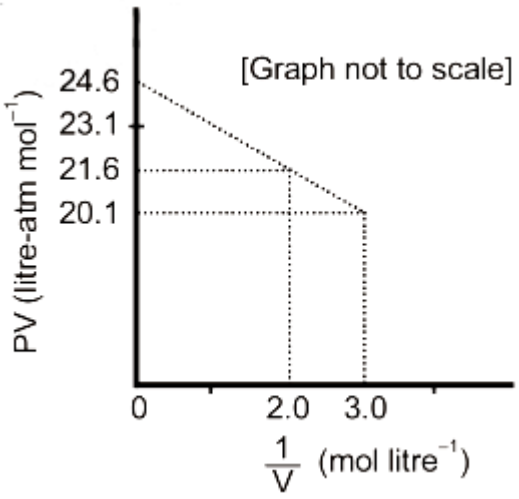
(B) $1.05\ \mathrm{atm}$.

(C) $2.05\ \mathrm{atm}$.

(D) $3.05\ \mathrm{atm}$.

Q.8

For one mole of a van der Waal's gas when $b = 0$ and $T = 300\ \mathrm{K}$, the PV vs. $1/\mathrm{N}$ plot is shown below. The value of the van der Waal's constant ' a ' (atm. litre mol^{-2}) is



(A) 1.0

(B) 4.5

(C) 1.5

(D) 3.0

Q.9

A balloon blown up with 1 mole of a gas, has a volume of $480\mathrm{ml}$ to $5^{\circ}\mathrm{C}$. If the balloon is filled to $\left(\frac{7}{8}\right)$ th of its maximum capacity, then which of the following options is/are correct?

(A) The balloon will burst at $30^{\circ}\mathrm{C}$

(B) The pressure of the gas inside the balloon at $5^{\circ}\mathrm{C}$ is $47.5\ \mathrm{atm}$

(C) The minimum temperature at which the balloon will burst is $44.71^{\circ}\mathrm{C}$

(D) The pressure of gas when balloon burst at minimum temperature is $50\ \mathrm{atm}$

Q.10

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An open vessel at $27^{\circ}\mathrm{C}$ is heated until $3/5^{\text{th}}$ of the air in it has been expelled. Assuming that the volume of the vessel remains constant. Choose the correct options -

(A) the air escaped out if vessel is heated to $900\ \mathrm{K}$ is $(\ 2/3\ )\ \mathrm{n}$ moles

(B) temperature at which half of the air escapes out is $600\ \mathrm{K}$

(C) the air escaped out if vessel is heated to $900\ \mathrm{K}$ is $(1/3)\mathrm{n}$ moles

(D) temperature at which half of the air escapes out is $400\ \mathrm{K}$

Q.11

$4.4\ \mathrm{g}\mathrm{CO}_2$ gas and 2.24 litre of H_2 gas at S.T.P. are taken in a 1 litre container at $27^{\circ}\mathrm{C}$. The total pressure of gases in container will be

(A) $4.926\ \mathrm{atm}$

(B) $3743.76\ \mathrm{mm}$ of Hg

(C) $1871.88\ \mathrm{mm}$ of Hg

(D) $2.463\ \mathrm{atm}$

Q.12

Paragraph:
$1\ \mathrm{mole}$ of CCl_4 vapours at $77^{\circ}\mathrm{C}$ occupies a volume of 35 litre. If vander waal's constant are $\mathrm{a} = 20.39\ \mathrm{L}^2\ \mathrm{atm}\ \mathrm{mol}^{-2}$ and $\mathrm{b} = 0.1383\ \mathrm{L}\ \mathrm{mol}^{-1}$.
Question:
Calculate Z under low pressure region -

(A) 0.98

(B) 0.90

(C) 0.84

(D) 0.80

Q.13

Paragraph:
$1\ \mathrm{mole}$ of CCl_4 vapours at $77^{\circ}\mathrm{C}$ occupies a volume of 35 litre. If vander waal's constant are $\mathrm{a} = 20.39\ \mathrm{L}^2\ \mathrm{atm}\ \mathrm{mol}^{-2}$ and $\mathrm{b} = 0.1383\ \mathrm{L}\ \mathrm{mol}^{-1}$.
Question:
Calculate Z under high pressure region -

(A) 1.20

(B) 1.30

(C) 1.25

(D) 1.004

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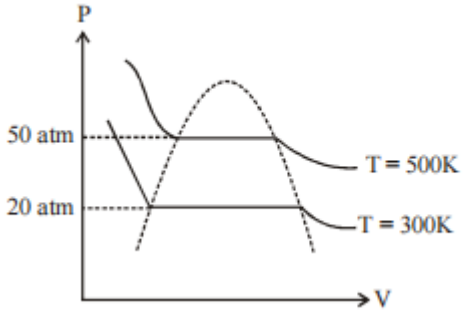
Q.14

Paragraph:

1 mole of CCl_4 vapours at $77^\circ C$ occupies a volume of 35 litre. If vander waal's constant are $a = 20.39\ L^2\ atm\ mol^{-2}$ and $b = 0.1383\ L\ mol^{-1}$.

Question:

For a real gas the P-V curve was experimentally plotted, and it had the following appearance with respect to liquification. Choose the correct statements.



(A) At $T = 500\ K, P = 40\ atm$, the state will be liquid

(B) At $T = 300\ K, P = 50$ atm, the state will be gas

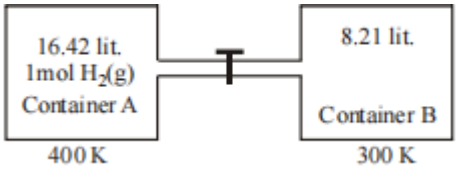
(C) At $T < 300\ K, P > 20$ atm, the state will be gas

(D) At $300\ K < T < 500\ K, P > 50$ atm, the state will be liquid

Q.15

Paragraph:

A container containing $H_2(g)$ was connected to another empty container as shown in diagram and maintained at given temperature.



Question:

The final pressure in each container is -

(A) 2 atm.

(B) $\frac{3}{5}$ atm

(C) $\frac{6}{5}$ atm

(D) $\frac{1}{6}$ atm

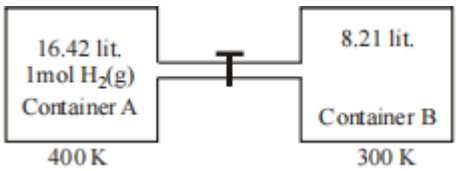
Q.16

Paragraph:

A container containing $H_2(g)$ was connected to another empty container as shown in diagram and maintained at

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given temperature.



Question:

The number of moles of H_2 in container B is -

(A) $\frac{3}{5}$

(B) $\frac{2}{5}$

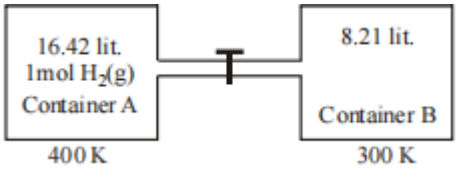
(C) $\frac{1}{2}$

(D) $\frac{4}{7}$

Q.17

Paragraph:

A container containing $H_2(g)$ was connected to another empty container as shown in diagram and maintained at given temperature.



Question:

The ratio of total number of collisions in container A to container B is -

(A) $\frac{3\sqrt{3}}{4}$

(B) $\frac{9}{16}$

(C) $\frac{3\sqrt{3}}{8}$

(D) None of these

Q.18

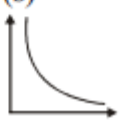
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Match the column -

(A) $1/V^2$ vs P for ideal gas at constant T and n. (p) 

(B) V Vs 1/T for ideal gas at constant P and n (q) 

(C) PT vs T^2 for ideal gas at constant T and n. (r) 

(D) V vs $1/P^2$ for ideal gas at constant T and n. (s) 

(A) A-r; B-s ; C- p ; D-q;

(B) A-p; B-r ; C- s ; D-s;

(C) A-s; B-r ; C-s ; D-s;

(D) A-s; B-p ; C- p ; D-s;

Q.19

A spherical glass bulb of radius $100\ \text{cm}$ contains a concentric rubber balloon that contains some N_2 gas and the remaining space in glass bulb contains $50\ \text{g}\ H_2$. In the given condition, radius of the rubber balloon was found to be $40\ \text{cm}$. Now, the seal of glass bulb was opened and $26\ \text{g}\ H_2(\text{g})$ was further added and resealed. Assuming constant temperature through out, if the radius of rubber balloon in the new conditions is $7x$, then what is x ?

Q.20

A sample of gas occupies 10 litre under a pressure of 1 atmosphere. What will be its volume (in litre) if the pressure is increased to 2 atmospheres? Assume that the temperature of the gas sample does not change.

Answers & Solutions

Q.1 Answer:

$5\mathring{A}$

Solution:

$Z = 1 + \frac{Pb}{RT}$ **At high pressure**

$\text{Slope} = \frac{b}{RT} = \frac{\pi}{492.6} \Rightarrow b = \frac{\pi}{492.6} \times 0.0821 \times 300$

$\text{and } b = \frac{4}{3}\pi r^3 \times 4\,N_A = \frac{\pi}{492.6} \times 0.0821 \times 300 \times 10^{-3}$

by solving this we get $\Rightarrow \quad r = 2.5\mathring{A}$ **or** $d = 5\mathring{A}$

Q.2 Answer:

Only (ii)

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Solution:

$M \propto \frac{n}{V(m^3)}$
$M \times V\left(m^3\right) =$ **constant rectangular hyperbola**

Q.3 Answer:

$4:1$

Solution:

$d_A = 2d_B$
$M_A = \dfrac{M_B}{2} \quad [T_1 = T_2]$
$P_A : P_B = ?$
$PV = nRT$
$\therefore P_A M_A = d_A RT \quad \ldots (i)$
and $P_B M_B = d_B RT \quad \ldots (ii)$
(i) $\div$ (ii)
$\therefore \dfrac{P_A M_A}{P_B M_B} = \dfrac{d_A}{d_B}$
$\therefore \dfrac{P_A}{P_B} = 2 \times 2$
$P_A : P_B = 4:1$

Q.4 Answer:

46.6

Solution:

The gaseous mixture contains $80\% O_2$ **and** 20% **gas.** $\therefore$ **Average molecular weight of mixture** $(\mathrm{M_{mix}}) = \frac{32\times 80 + 20\times \mathrm{m}}{100} \quad$ **... (i)**

Now for diffusion of gaseous mixture and pure O_2

$$\frac{\mathrm{r_{O_2}}}{\mathrm{r_m}} = \sqrt{\frac{\mathrm{M_{mix}}}{\mathrm{M_{O_2}}}} \text{ or } \frac{\mathrm{V_{O_2}}}{\mathrm{V_{mix}}} \times \frac{\mathrm{t_{mix}}}{\mathrm{t_{O_2}}} = \sqrt{\frac{\mathrm{M_{mix}}}{32}}$$

$$\text{or} \quad \frac{1}{224} \times \frac{234}{1} = \sqrt{\frac{\mathrm{M_{mix}}}{32}}$$

$$\therefore \quad \mathrm{M_{mix}} = 34.92$$

By (i) and (ii) mol weight of gas $(\mathrm{m}) = 46.6$**.**

Q.5 Answer:

3.1 litre

Solution:

At constant volume,
$\frac{P_1}{T_1} = \frac{P_2}{T_2}$

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or $\frac{3}{290} = \frac{P_2}{320}$

or $P_2 = \frac{320 \times 3}{290} = 3.31 \text{ atm}$

Pressure to be decreased in tyre

$= 3.31 - 3.0 = 0.31 \text{ atm}$

Let the volume of the gas to be taken out at 1 atmospheric pressure be V. As the temperature remains constant, i.e., $47^\circ \mathrm{C}$**, Boyle's law can be applied to determine** V**.**

$$1 \times \mathrm{V} = 0.31 \times 10$$

or $\mathrm{V} = 3.1 \text{ litre}$

Q.6 Answer:

$1/6$

Solution:

Suppose initially, pressure of $\mathrm{C_6H_6(\ g)} = \mathrm{p_1}$ **mm and that of** $\mathrm{H_2(\ g)} = \mathrm{p_2}$ **mm**

$\therefore \quad \mathrm{p_1 + p_2} = 80 \text{ mm}$ **...(i)**

After the reaction,

pressure of $\mathrm{C_6H_6(\ g)} = 0$

(as all has reacted)

pressure of $\mathrm{H_2(\ g)} = \mathrm{p_2 - 3p_1}$

pressure of $\mathrm{C_6H_{12}(\ g)} = \mathrm{p_1}$

Total pressure $= \mathrm{p_2 - 3p_1 + p_1} = 40 \text{ mm}$

or $\mathrm{p_2 - 2p_1} = 40 \text{ mm}$ **...(ii)**

Solving (i) and (ii), we get \& applying Fraction of $\mathrm{C_6H_6}$ **by volume = fraction of moles Fraction of pressure** $= \frac{40}{3 \times 80} = \frac{1}{6}$.

Q.7 Answer:

4.05 atm.

Solution:

Let us take moles of $\mathrm{C_2H_4}$ **and** $\mathrm{O_2}$ **are and respectively.**

$\mathrm{n} = \frac{\mathrm{PV}}{\mathrm{RT}} \quad \therefore \mathrm{n_{C_2H_4}} = \frac{0.5}{\mathrm{RT}}; \mathrm{n_{O_2}} = \frac{5 \times 0.5}{\mathrm{RT}} = \frac{2.5}{\mathrm{RT}}$

$\mathrm{C_2H_4 + 3O_2 = 2CO_2 + 2H_2O}$

So after the reaction moles of $\mathrm{O_2}$ **left** $= \frac{2.5}{\mathrm{RT}} - \frac{1.5}{\mathrm{RT}} = \frac{1}{\mathrm{RT}}$

Also moles of $\mathrm{CO_2}$ **produced** $= \frac{2 \times 0.5}{\mathrm{RT}} = \frac{1}{\mathrm{RT}}$

Total number of moles $= \frac{1}{\mathrm{RT}} + \frac{1}{\mathrm{RT}} = \frac{2}{\mathrm{RT}}$

Pressure due to $\mathrm{CO_2}$ **and** $\mathrm{O_2}$ $\mathrm{P'} = \frac{2}{\mathrm{RT}} \times \frac{\mathrm{RT}}{0.5} \text{ atm} = 4 \text{ atm}$ **(converting aqueous tension in atm)**

Total pressure $\mathrm{P} = \mathrm{P'}$ **+ aqu. tension** $= 4 + \frac{38}{7}\ 4 + 0.05 = 4.05 \text{ atm.}$

Q.8 Answer:

1.5

Solution:

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$\left(\mathrm{P}+\frac{\mathrm{a}}{\mathrm{V}^2}\right) \mathrm{V}=\mathrm{RT}$

$$PV+\frac{a}{V}=RT$$

$$PV=RT-\frac{a}{V}$$

So, Plot of (PV) vs $\left(\frac{1}{V}\right)$ has slope $-a$.

$$\therefore \text{ slope } =\frac{20.1-24.6}{3}=-1.5$$

$$\therefore a=1.5 \text{ atmL }^2\text{mol}^{-2}$$

Q.9 Answer:

The pressure of the gas inside the balloon at $5^{\circ}\mathrm{C}$ is 47.5 atm

Solution:

Maximum capacity or volumes of balloon $=\frac{8}{7} \times 480=548.57\mathrm{ml}$

Also, $\mathrm{V}_1=480\mathrm{ml}, \mathrm{T}_1=278 \mathrm{~K}, \mathrm{n}=1$ **mole**

(A) The balloon will burst at a minimum temperature $\left(T_2\right)$ **when volume becomes** $548.57\mathrm{ml}$.

Using Charle's law

$$\therefore \frac{V_1}{T_1}=\frac{V_2}{T_2}$$

$$\frac{480}{278}=\frac{548.57}{T_2}$$

$$\therefore T_2=317.71 \text{ K}$$

$$\text{or } T_2=44.71^{\circ}\mathrm{C}$$

Hence (A) **is incorrect but** (C) **is correct.**

(B) Pressure of the gas at $5^{\circ}\mathrm{C}$ **having 1 mole and** $\mathrm{V}=480\mathrm{ml}$

$$\mathrm{PV}=\mathrm{nRT}$$

$$\therefore \mathrm{P} \times \frac{480}{1000}=1 \times 0.0821 \times 278$$

$$\therefore \mathrm{P}=47.5 \text{ atm}$$

Hence (B) **is also correct.**

(D) Since V increase with increase in temperature from $5^{\circ}\mathrm{C}$ **to** $44.71^{\circ}\mathrm{C}$**, at which balloon bursts and therefore pressure remains constant. Thus pressure of gas inside the balloon when it bursts is** 47.5 atm.

Hence (D) is incorrect.

Hence (b, c) **are correct.**

Q.10 Answer:

the air escaped out if vessel is heated to 900 K is $(2/3)$ n moles

Solution:

One should clearly note the fact that on heating a gas in a vessel, there are the number of moles of gas which go out, the volume of vessel remains constant.

Let initial no. of moles of gas at 300 K **be ' n '. On heating** $3/5$ **moles of air escape out at temperature** T.

$\therefore$ **Moles of air left at temperature** $T=\left(n-\frac{3}{5} n\right)=\frac{2 n}{5}$

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(A) On heating vessel to $900\ \mathrm{K}$, let $\mathrm{n_1}$ moles be left,

$$\mathrm{n_1\ T_1 = n_2\ T_2 \Rightarrow n_1 \times 900 = 300 \times n \Rightarrow n_1 = n/3}$$

$\therefore$ moles escaped out $= \mathrm{n - \frac{n}{3} = \frac{2}{3}n}$ moles

(B) Let $\mathrm{n/2}$ moles escape out at temperature T, then

$$\mathrm{n_1\ T_1 = n_2\ T_2 \Rightarrow \frac{n}{2} \times T = n \times 300 \Rightarrow T = 600\ K}$$

Q.11 Answer:

$4.926\ \mathrm{atm}$

Solution:

$PV = nRT$

Total pressure $\mathrm{(P_T) = P_1 + P_2}$

or, $\mathrm{P_T = \frac{(n_1+n_2)RT}{V}}$ $\quad[\mathrm{P_T \rightarrow}$ **Total pressure]**

$\mathrm{n_1\ (CO_2) = 0.1\ mole}$

$\mathrm{n_2\ (H_2) = \frac{2.24}{22.4} = 0.1\ mole}$

$\therefore \mathrm{P_T = \frac{(0.1+0.1)\times 0.0821 \times 300}{1}}$

$\mathrm{P_T = 4.926\ atm}$

$\mathrm{1\ atm = 760\ mm - Hg}$

$\therefore \mathrm{P_T = 3743.76\ mm - Hg}$

Q.12 Answer:

0.98

Solution:

$\mathrm{\left(P + \frac{a}{V_m^2}\right) V_m = RT}$ **or** $\mathrm{PV_m + \frac{a}{V_m} = RT}$

or $\mathrm{\frac{PV_m}{RT} + \frac{a}{RTV_m} = 1}$ **or** $\mathrm{Z = 1 - \frac{a}{RTV_m} = 1 - \frac{20.39}{0.0821 \times 350 \times 35} = 0.98}$

Q.13 Answer:

1.004

Solution:

$\mathrm{P\ (V_m - b) = RT}$ **or** $\quad \mathrm{PV_m - Pb = RT}$

or $\quad Z = 1 + \frac{\mathrm{Pb}}{RT}$**...(1)**

or $\quad P = \frac{RT}{V_m - b}$**...(2)**

From eq. (1) and eq. (2)

$$Z = 1 + \frac{b}{(V_m - b)} = 1 + \frac{0.1383}{35 - 0.1383} = 1.004$$

Q.14 Answer:

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At $300\ \mathrm{K} < \mathrm{T} < 500\ \mathrm{K}, \mathrm{P} > 50$ atm, the state will be liquid

Solution:

Inside the dome: State is liquid.

Q.15 Answer:

$\frac{6}{5}$ atm

Solution:

$\mathrm{P} =$ **final pressure**

$\mathrm{P} = \frac{\mathrm{n_A RT_A}}{\mathrm{V_A}} = \frac{(1-\mathrm{n_A})\mathrm{RT_B}}{\mathrm{V_B}} \Rightarrow \frac{\mathrm{n_A} \times 400}{16.42} = \frac{(1-\mathrm{n_A}) \times 300}{8.21} \Rightarrow \mathrm{n_A} = \frac{3}{5}, \mathrm{n_B} = \frac{2}{5}$

$\mathrm{n_t} = \mathrm{n_A} + \mathrm{n_B} = 1$ mole

$\mathrm{n_B} = (1 - \mathrm{n_A})$

$\mathrm{P} = \frac{\mathrm{n_A RT_A}}{\mathrm{V_A}} = \frac{(1-\mathrm{n_A})\mathrm{RT_B}}{\mathrm{V_B}} \Rightarrow \mathrm{P} = \frac{3 \times 0.0821 \times 400}{5 \times 16.42\ \text{atm} = \frac{6}{5}\ \text{atm}}$

$[\mathrm{n_{H_2}}]_\mathrm{B} = \frac{\mathrm{PV}}{\mathrm{RT}} = \frac{6/5 \times 8.21}{0.0821 \times 300} = \frac{2}{5}$

Q.16 Answer:

$\frac{2}{5}$

Solution:

$[\mathrm{n_{H_2}}]_\mathrm{B} = \frac{\mathrm{PV}}{\mathrm{RT}} = \frac{6/5 \times 8.21}{0.0821 \times 300} = \frac{2}{5}$

Q.17 Answer:

$\frac{3\sqrt{3}}{4}$

Solution:

Total number of collision in container

$\frac{Z_\mathrm{A}}{Z_\mathrm{B}} = \frac{\sqrt{\frac{2\mathrm{RT_A}}{\mathrm{M}}}}{\sqrt{\frac{2\mathrm{RT_B}}{\mathrm{M}}}} \times \frac{\left[\frac{\mathrm{PN_A}}{\mathrm{RT}}\right]^2}{\left[\frac{\mathrm{PN_B}}{\mathrm{RT}}\right]^2} \times \frac{\mathrm{V_A}}{\mathrm{V_B}} = \frac{\sqrt{400}}{\sqrt{300}} \times \frac{\left[\frac{1}{400}\right]^2}{\left[\frac{1}{300}\right]^2} \times \left[\frac{2}{1}\right]$

$= \sqrt{\frac{4}{3}} \times \frac{\frac{1}{16}}{\frac{1}{9}} \times \frac{2}{1} = \sqrt{\frac{4}{3}} \times \frac{9}{16} \times \frac{4}{1} = \frac{3\sqrt{3}}{4}$

Q.18 Answer:

A-r; B-s ; C- p ; D-q;

Solution:

((A)-r,(B)-s,(C)-p,(D)-q)

$\mathrm{y} = \frac{1}{\mathrm{V}^2}$ or $\sqrt{\mathrm{y}} = \frac{1}{\mathrm{V}}$

$\mathrm{P} = \mathrm{x}$ and $\mathrm{P} = \frac{\text{Constant}}{\mathrm{V}}$

$(\mathrm{A})\ x = (k)\sqrt{y} \Rightarrow y = k^1 x^2$

(B) $\mathrm{V} = \mathrm{kT}, \mathrm{y} = \mathrm{V}\ \&\ \frac{1}{\mathrm{T}} = \mathrm{x} \therefore \mathrm{y} = \frac{\mathrm{k}}{\mathrm{x}}$

(C) $\mathrm{p} = \mathrm{kT}; \mathrm{PT} = \mathrm{kT}^2$ **or** $\mathrm{y} = \mathrm{kx}$

(D) $\mathrm{v} = \frac{\mathrm{c}}{\mathrm{p}} \Rightarrow \mathrm{y} = \mathrm{c}\sqrt{\mathrm{x}}; \mathrm{y}^2 = \mathrm{cx}$

Q.19 Answer:

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Solution:

In case-I : $\mathrm{P_{H_2}} = \mathrm{P_{N_2}}$

$$\Rightarrow \frac{\mathrm{V_{H_2}}}{\mathrm{n_{H_2}}} = \frac{\mathrm{V_{N_2}}}{\mathrm{n_{N_2}}}$$

$$\Rightarrow \frac{\frac{4}{3}\pi\left(100^3 - 40^3\right)}{25} = \frac{\frac{4}{3}\pi 40^3}{\mathrm{n_{H_2}}}$$

$$\Rightarrow \mathrm{n_{H_2}} = 1.7 \text{ mole}$$

In case-II : Some extra $\mathrm{H_2}$ **was introduced in glass bulb to the rubber balloon will contract Ultimately**

$\mathrm{P_{H_2}} = \mathrm{P_{N_2}}$

$\Rightarrow \frac{\mathrm{V_{H_2}}}{\mathrm{n_{H_2}}} = \frac{\mathrm{V_{N_2}}}{\mathrm{n_{N_2}}}$

$\Rightarrow \frac{\frac{4}{3}\pi\left(100^3 - x^3\right)}{38} = \frac{\frac{4}{3}\pi r^3}{1.7}$

$\Rightarrow x = 35.08 \approx 35 = 7 \times 5$

$\therefore x = 5$

Q.20 Answer:

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Solution:

Mass and temperature is constant hence we can use Boyle's law $\mathrm{P_1\ V_1 = P_2\ V_2}$

Given $P_1 = 1 \text{ atm}, V_1 = 10$ **litre,** $P_2 = 2 \text{ atm}, V_2 = ?$

$$1 \times 10 = 2 \times \mathrm{V_2} \Rightarrow \mathrm{V_2} = 5 \text{ litre}$$

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